

Infrared Thermography Temperature Dataset Analysis

K.A.S.M.Abeyrathne

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Department of Statistics and Computer Science

University of Peradeniya

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1 Introduction

The Infrared Thermography Temperature Dataset contains temperature measurements extracted from thermal images of human subjects, along with oral temperature readings used as the target variable. Each record represents an individual observation collected under different environmental conditions such as ambient temperature, humidity, and distance from the sensor.

The dataset includes 1020 instances and 33 features consisting of demographic variables (gender, age, ethnicity), environmental factors, and detailed facial thermal readings from multiple regions of interest. The main goal of this dataset is to model and understand how thermal image features and environmental conditions relate to human body temperature, particularly oral temperature.

1.1 Objectives

The main objective of this study is to analyze the structure and relationships within the dataset using multivariate statistical techniques. Specifically, the study aims to reduce dimensionality, identify hidden patterns among thermal features, and explore relationships between different groups of variables.

Another key objective is to determine how well environmental conditions and facial thermal readings can collectively explain variations in oral temperature.

1.2 Major Question

Can oral temperature be effectively explained or approximated using environmental conditions and thermal image features, while reducing the complexity of the dataset through multivariate statistical techniques?

1.3 Purpose

The purpose of this study is to analyze the Infrared Thermography Temperature Dataset to understand its underlying structure and relationships using multivariate statistical techniques. Since the dataset contains 33 mixed-type features including environmental variables, demographic information, and thermal image temperature readings, direct interpretation is complex and high-dimensional.

To address this, the following methods are applied:

- Principal Component Analysis (PCA) to reduce dimensionality while retaining most of the variation in the dataset.

- Factor Analysis (FA) to identify hidden latent structures that explain correlations among variables.
- Canonical Correlation Analysis (CCA) to study the relationship between two groups of variables, such as environmental/demographic features and thermal image temperature features.

Overall, this study aims to extract meaningful patterns, reduce redundancy, and improve interpretability of the dataset for further analytical and predictive modeling.

2 Methodology

2.1 Dataset Description

The Infrared Thermography Temperature Dataset contains temperatures read from various locations of infrared images of patients, along with oral temperature measurements for each individual. The dataset includes 33 features such as gender, age, ethnicity, ambient temperature, humidity, distance, and multiple thermal image-based temperature readings.

The dataset is designed for a regression task, where the goal is to predict oral temperature using environmental and thermal image features.

Table 1: Summary of Dataset Information

Property	Details
Dataset Name	Infrared Thermography Temperature Dataset
Subject Area	Health and Medicine
Task Type	Regression
Number of Instances	1020
Number of Features	33
Feature Types	Continuous and Categorical
Target Variable	Oral Temperature
Data Type	Tabular

```

...   Gender   Age           Ethnicity T_atm Humidity Distance \
0    Male  41-50           White    24.0    28.0     0.8
1  Female  31-40  Black or African-American  24.0    26.0     0.8
2  Female  21-30           White    24.0    26.0     0.8
3  Female  21-30  Black or African-American  24.0    27.0     0.8
4    Male  18-20           White    24.0    27.0     0.8

      T_offset1 Max1R13_1 Max1L13_1 aveAllR13_1 ... T_FHCC1 T_FHRC1 \
0      0.7025    35.0300    35.3775    34.4000 ...   33.5775   33.4775
1      0.7800    34.5500    34.5200    33.9300 ...   34.0325   34.0550
2      0.8625    35.6525    35.5175    34.2775 ...   34.9000   34.8275
3      0.9300    35.2225    35.6125    34.3850 ...   34.4400   34.4225
4      0.8950    35.5450    35.6650    34.9100 ...   35.0900   35.1600

      T_FHLC1 T_FHBC1 T_FHTC1 T_FH_Max1 T_FHC_Max1 T_Max1 T_OR1 \
0   33.3725  33.4925  33.0025  34.5300    34.0075  35.6925  35.6350
1   33.6775  33.9700  34.0025  34.6825    34.6600  35.1750  35.0925
2   34.6475  34.8200  34.6700  35.3450    35.2225  35.9125  35.8600
3   34.6550  34.3025  34.9175  35.6025    35.3150  35.7200  34.9650
4   34.3975  34.6700  33.8275  35.4175    35.3725  35.8950  35.5875

      T_OR_Max1
0      35.6525
1      35.1075
2      35.8850
3      34.9825
4      35.6175

[5 rows x 33 columns]
(1020, 33)

```

Figure 1: Sample view of the Infrared Thermography Temperature Dataset

2.2 Data Preprocessing

Data preprocessing was performed to ensure the dataset is clean and suitable for analysis. The following steps were carried out:

- **Handling Missing Values:** Missing values were checked across all variables. Only the *Distance* variable had 2 missing values, which were replaced using mean imputation.
- **Encoding Categorical Variables:** Categorical features such as Gender, Age, and Ethnicity were converted into numerical form using label encoding.
- **Outlier Removal:** Outliers were detected and removed using the Isolation Forest method with a contamination level of 5%. This reduced the dataset size from 1020 to 969 observations.
- **Visualization:** Boxplots were used to visualize the data before and after outlier removal, confirming the reduction of extreme values.
- **Feature Scaling:** All variables were standardized using the StandardScaler method, transforming the data to have a mean of zero and a standard deviation of one.

The cleaned and standardized dataset was then used for further analysis.

2.3 Methods Employed

2.3.1 Principal Component Analysis (PCA)

Principal Component Analysis was applied to reduce the dimensionality of the dataset and identify key patterns.

- **Scree Plot:**

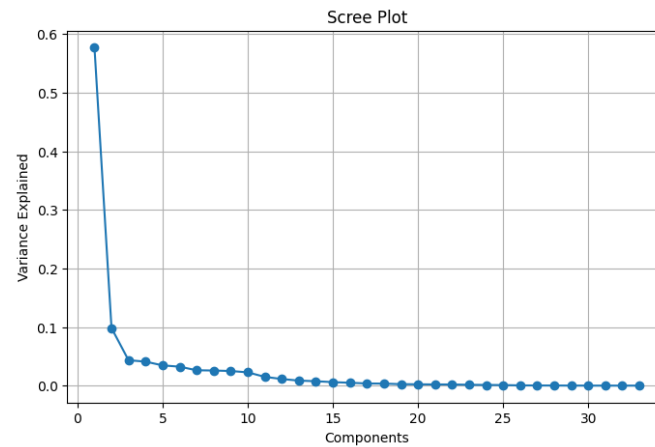


Figure 2: Scree Plot of Explained Variance

- Shows the variance explained by each principal component.
- A clear elbow appears around PC3–PC4.
- After PC4, additional components contribute very little variance.
- Indicates that a small number of components is sufficient.

- **Cumulative Explained Variance:**

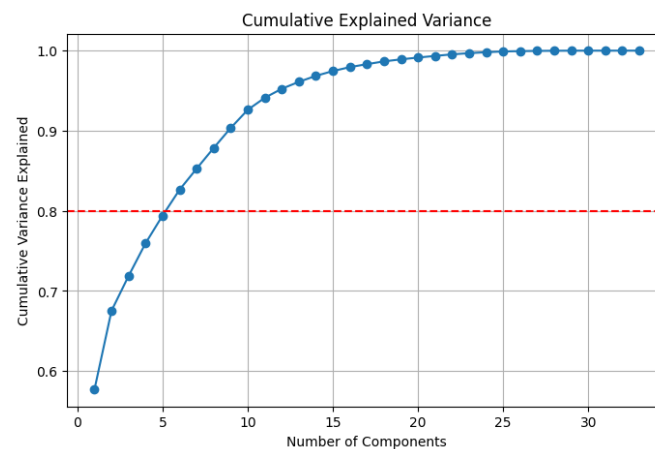


Figure 3: Cumulative Explained Variance

- Shows how total variance accumulates across components.
- Most of the variance is captured within the first few components.
- Around 4–6 components explain the majority of variability.
- Confirms effective dimensionality reduction.

• PCA Loading Heatmap:

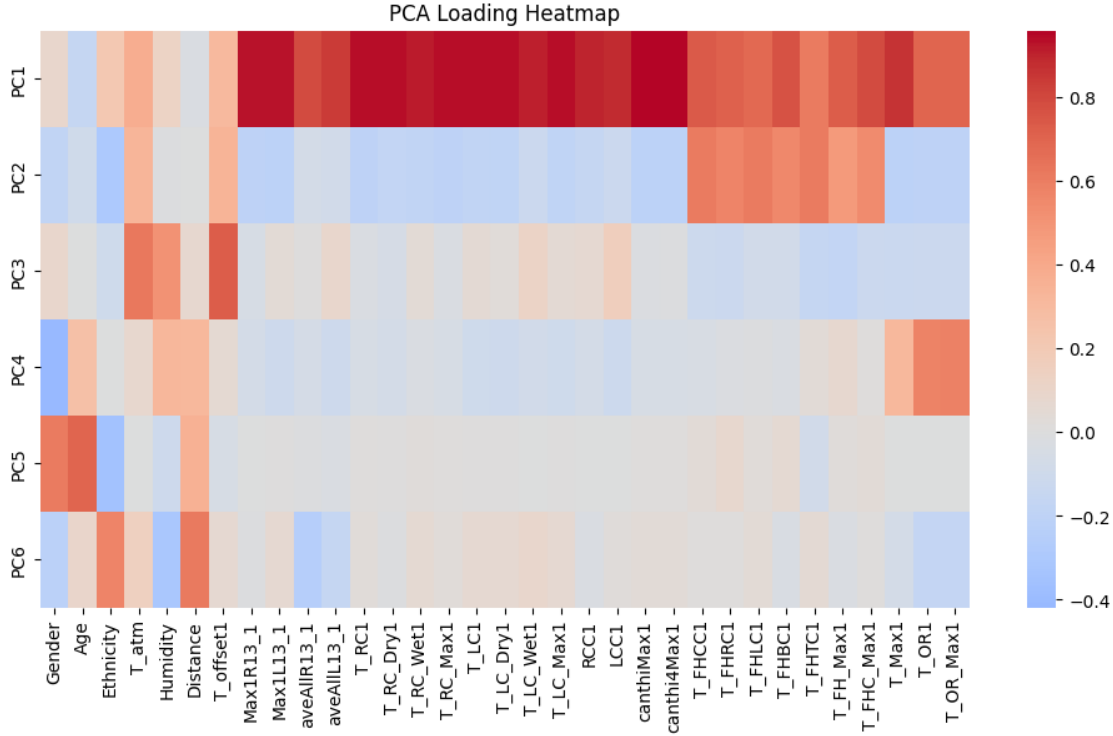


Figure 4: Heatmap of PCA Loadings (Top 6 Components)

- Displays how strongly each variable contributes to components.
- High values indicate strong influence.
- Helps identify groups of related variables.

• Component Interpretation:

- **PC1:** Dominated by canthiMax1, canthi4Max1, T_LC1, T_RC1, T_RC_Max1. Represents overall facial temperature (especially eye region).
- **PC2:** Influenced by forehead variables (T_FHLC1, T_FHTC1, T_FHCC1, T_FHRC1, T_FHBC1). Captures forehead temperature patterns.

- **PC3:** Driven by T_offset1, T_atm, and Humidity. Represents environmental conditions.
- **PC4:** Influenced by T_OR1, T_OR_Max1, Gender, and Humidity. Reflects mouth temperature and demographic effects.
- **PC5:** Dominated by Age, Gender, Distance, and Ethnicity. Represents demographic characteristics.
- **PC6:** Influenced by Distance and Ethnicity. Shows mixed environmental and demographic variation.

- **Overall Insight:**

- A small number of components can represent the dataset effectively.
- Thermal, environmental, and demographic features are clearly separated.
- PCA simplifies the dataset while retaining most of the important information.

2.3.2 Factor Analysis (FA)

Factor Analysis was applied to identify underlying latent structures in the dataset.

- **Suitability Tests for Factor Analysis:**

- KMO (Kaiser-Meyer-Olkin) Score = 0.938
- Bartlett's Test p-value = 0.00000
- Since $KMO > 0.9$, the sampling adequacy is excellent.
- The significant Bartlett's test ($p < 0.05$) indicates that variables are sufficiently correlated.
- Therefore, Factor Analysis is appropriate for this dataset.

- **Factor Loadings Heatmap:**

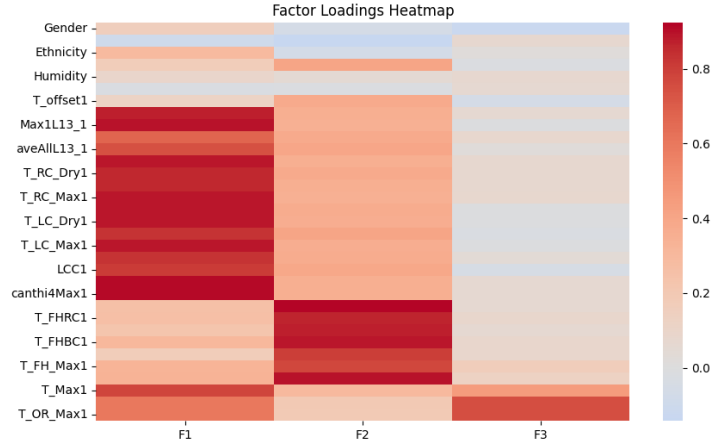


Figure 5: Heatmap of Factor Loadings

- The heatmap shows how strongly each variable loads onto each factor.
- High values (close to 1 or -1) indicate strong relationships.
- Helps visualize grouping of variables into factors.

• **Factor Interpretation:**

– **Factor 1 (F1):**

- * Strong loadings observed in thermal features such as canthiMax1 (0.91), canthi4Max1 (0.91), T_RC1 (0.88), T_LC1 (0.89), and related variables.
- * Many facial temperature variables (eye and cheek regions) load highly on this factor.
- * This factor represents overall facial thermal intensity.

– **Factor 2 (F2):**

- * High loadings seen in forehead-related variables such as T_FHCC1 (0.92), T_FHRC1 (0.87), T_FHLC1 (0.87), T_FHBC1 (0.89), and T_FHC_Max1 (0.89).
- * Environmental variables like T_atm and T_offset1 also show moderate contribution.
- * This factor represents forehead temperature variation.

– **Factor 3 (F3):**

- * High loadings observed in mouth-related variables such as T_OR1 (0.75) and T_OR_Max1 (0.75).
- * T_Max1 also contributes moderately.
- * This factor captures mouth temperature characteristics.

- **Overall Insight:**

- The dataset can be summarized into three main latent factors.
- Factor 1 represents overall facial temperature.
- Factor 2 represents forehead temperature patterns.
- Factor 3 represents mouth temperature features.
- This reduces complexity and improves interpretability of the dataset.

2.3.3 Canonical Correlation Analysis (CCA)

Canonical Correlation Analysis (CCA) was applied to examine the relationship between two sets of variables.

- **Variable Sets:**

- Set 1: Environment and Demographic variables (Gender, Age, Ethnicity, T_atm, Humidity, Distance, T_offset1)
- Set 2: Thermal features (facial temperature measurements)

- **Canonical Correlations:**

- Component 1: 0.5819
- Component 2: 0.5204

- **Bar Plot of Canonical Correlations:**

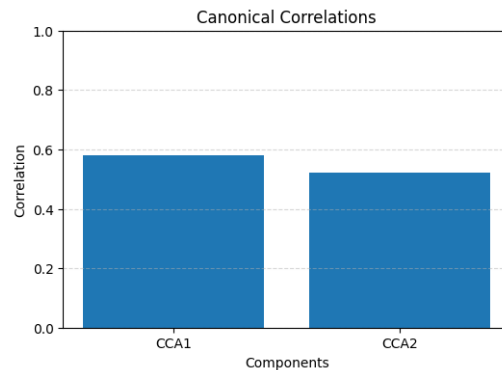


Figure 6: Canonical Correlations for Each Component

- The bar chart shows the strength of correlation for each canonical component.

- Both components show moderate correlation between the two variable sets.

- **Scatter Plot of Canonical Variables:**

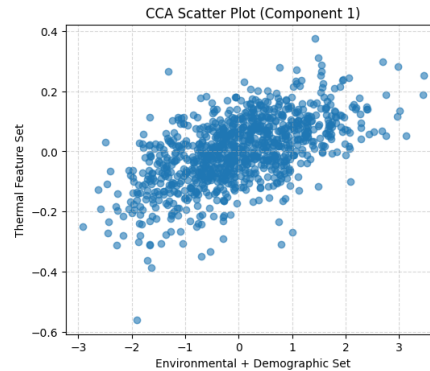


Figure 7: Scatter Plot of Canonical Variables

- The scatter plot shows the relationship between the first pair of canonical variables.
- Two colors represent the two sets of variables.
- A positive pattern indicates that as one set increases, the other also tends to increase.

- **Correlation Matrix of Canonical Components:**

- The matrix shows correlations between canonical variables from both sets.
- Strong relationships are observed between corresponding pairs:
 - * $X1_CC1$ and $X2_CC1 = 0.5819$
 - * $X1_CC2$ and $X2_CC2 = 0.5204$
- Other cross-correlations are close to zero, indicating independence between components.

- **Interpretation:**

- The first canonical correlation (0.5819) indicates a moderate relationship between environmental/demographic variables and thermal features.
- The second canonical correlation (0.5204) also shows a meaningful but slightly weaker relationship.
- These results suggest that environmental conditions and subject characteristics influence thermal measurements.

- However, the relationships are not very strong, meaning other factors may also affect thermal readings.
- Overall, CCA shows that there is a noticeable but moderate connection between the two variable sets.

3 Results Summary

This study uses the Infrared Thermography Temperature Dataset to analyze temperature patterns from thermal images and environmental conditions. The dataset contains 1020 observations and 33 features, including demographic, environmental, and thermal variables.

Three methods were used: Principal Component Analysis (PCA), Factor Analysis (FA), and Canonical Correlation Analysis (CCA). PCA helped reduce the number of variables while keeping most of the information. FA grouped related variables into factors, making the data easier to understand. CCA was used to study the relationship between environmental variables and thermal measurements.

The results showed that most of the variation in the data comes from thermal features. Also, there is a moderate relationship between environmental conditions and thermal readings.

4 Conclusion

The analysis shows that the dataset can be simplified without losing important information. PCA reduced the dimensions, and FA identified meaningful groups of variables.

CCA results show that environmental factors like temperature and humidity have some influence on thermal readings, but the relationship is not very strong.

Overall, the dataset is suitable for predicting oral temperature using thermal and environmental data.

5 Limitations

- The methods assume linear relationships, which may not always be true
- Interpretation of components and factors is sometimes difficult
- High correlation between variables can affect results
- CCA showed only moderate relationships

6 Recommendations

- Use machine learning models for better prediction
- Reduce highly correlated features
- Apply cross-validation for better accuracy
- Add more relevant variables for better analysis

7 References

References

- [1] Wang, Q., Zhou, Y., Ghassemi, P., McBride, D., Casamento, J., & Pfefer, T. (2023). *Infrared thermography temperature dataset*. UCI Machine Learning Repository. <https://archive.ics.uci.edu/dataset/925/infrared+thermography+temperature+dataset>
- [2] Jolliffe, I. T., & Cadima, J. (2016). Principal component analysis: A review and recent developments. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 374(2065), 20150202. <https://doi.org/10.1098/rsta.2015.0202>
- [3] OpenAI. (2024). *ChatGPT* (GPT-4/5 version) [Large language model]. <https://chat.openai.com/>

8 Appendices

• Library Installation and Importing Packages

The following code installs required libraries and imports all necessary packages for data analysis and visualization.

```
1 !pip install factor_analyzer
2 !pip install ucimlrepo
3
4 import pandas as pd
5 import numpy as np
6 import seaborn as sns
7 import matplotlib.pyplot as plt
8
9 from sklearn.preprocessing import StandardScaler
```

```

10 from sklearn.ensemble import IsolationForest
11 from sklearn.decomposition import PCA
12 from sklearn.cross_decomposition import CCA
13 from factor_analyzer import FactorAnalyzer
14 from ucimlrepo import fetch_ucirepo

```

- **Loading the Dataset**

This code fetches the Infrared Thermography dataset from the UCI repository and extracts features and target variables.

```

1 infrared = fetch_ucirepo(id=925)
2
3 X = infrared.data.features.copy()
4 y = infrared.data.targets.copy()
5
6 print(X.head())
7 print(X.shape)

```

- **Handling Missing Values**

This code checks for missing values and fills them using the mean of numerical columns.

```

1 print("\nMissing values:\n", X.isnull().sum())
2
3 X = X.fillna(X.mean(numeric_only=True))

```

- **Label Encoding**

This code converts categorical variables such as Gender, Age, and Ethnicity into numerical form.

```

1 from sklearn.preprocessing import LabelEncoder
2
3 df = X.copy()
4
5 le_gender = LabelEncoder()
6 df['Gender'] = le_gender.fit_transform(df['Gender'])
7
8 le_age = LabelEncoder()
9 df['Age'] = le_age.fit_transform(df['Age'])
10
11 le_ethnicity = LabelEncoder()

```

```

12 df['Ethnicity'] = le_ethnicity.fit_transform(df['Ethnicity'])
13
14 y = y.iloc[:, 0]

```

• Boxplot Before Outlier Removal

The following code visualizes the dataset using a boxplot to identify potential outliers before applying any outlier detection method.

```

1 plt.figure(figsize=(12,6))
2 sns.boxplot(data=df)
3 plt.title("Before Outlier Removal")
4 plt.xticks(rotation=90)
5 plt.show()

```

• Outlier Detection and Removal

This code applies the Isolation Forest algorithm to detect and remove outliers from the dataset.

```

1 iso = IsolationForest(contamination=0.05, random_state=42)
2
3 outliers = iso.fit_predict(df)
4 mask = outliers != -1
5
6 X_clean = df[mask].reset_index(drop=True)
7 y_clean = y.iloc[mask].reset_index(drop=True)
8
9 print("Original shape:", df.shape)
10 print("Clean shape:", X_clean.shape)

```

• Cleaned Dataset Overview

This code displays the first few rows, structure, and shape of the cleaned dataset after removing outliers.

```

1 print("\n==== CLEANED DATA HEAD =====")
2 print(X_clean.head())
3
4 print("\n==== CLEANED DATA INFO =====")
5 print(X_clean.info())
6
7 print("\n==== CLEANED SHAPE =====")

```

```
8 print(X_clean.shape)
```

- **Boxplot After Outlier Removal**

This code visualizes the dataset again using a boxplot to confirm that outliers have been removed.

```
1 plt.figure(figsize=(12,6))
2 sns.boxplot(data=X_clean)
3 plt.title("After Outlier Removal")
4 plt.xticks(rotation=90)
5 plt.show()
```

- **Data Standardization**

This code standardizes the cleaned dataset using StandardScaler to ensure all features are on the same scale.

```
1 from sklearn.preprocessing import StandardScaler
2
3 scaler = StandardScaler()
4 X_scaled = scaler.fit_transform(X_clean)
5
6 X_scaled_df = pd.DataFrame(X_scaled, columns=X_clean.columns)
```

- **Principal Component Analysis (PCA)**

This code applies PCA to reduce dimensionality and transform the standardized dataset into principal components.

```
1 from sklearn.decomposition import PCA
2
3 pca = PCA()
4 X_pca = pca.fit_transform(X_scaled)
```

- **Scree Plot**

This code generates a scree plot to visualize the variance explained by each principal component.

```
1 plt.figure(figsize=(8,5))
2 plt.plot(range(1, len(pca.explained_variance_ratio_)+1),
3          pca.explained_variance_ratio_,
4          marker='o')
```

```

5
6 plt.title("Scree_Plot")
7 plt.xlabel("Components")
8 plt.ylabel("Variance_Explained")
9 plt.grid()
10 plt.show()

```

• Selecting Components Based on Variance

This code selects the number of principal components required to retain at least 80% of the total variance.

```

1 pca_80 = PCA(n_components=0.80)
2 X_pca_80 = pca_80.fit_transform(X_scaled)
3
4 print("Selected_components:", pca_80.n_components_)

```

• PCA Loading Heatmap

This code computes PCA loadings and visualizes them using a heatmap.

```

1 loadings = pca_80.components_.T * np.sqrt(pca_80.
2     explained_variance_)
3
4 loadings_df = pd.DataFrame(
5     loadings,
6     index=X_clean.columns,
7     columns=[f"PC{i+1}" for i in range(loadings.shape[1])]
8 )
9
10 plt.figure(figsize=(12,6))
11 sns.heatmap(loadings_df.T, cmap="coolwarm", center=0)
12 plt.title("PCA_Loading_Heatmap")
13 plt.show()

```

• Cumulative Explained Variance Plot

This code shows how variance accumulates as more principal components are added.

```

1 pca = PCA()
2 X_pca = pca.fit_transform(X_scaled)
3
4 cumulative_variance = np.cumsum(pca.explained_variance_ratio_)

```

```

5
6 plt.figure(figsize=(8,5))
7 plt.plot(range(1, len(cumulative_variance)+1),
8          cumulative_variance,
9          marker='o')
10
11 plt.axhline(y=0.80, color='r', linestyle='--')
12 plt.title("Cumulative Explained Variance")
13 plt.xlabel("Number of Components")
14 plt.ylabel("Cumulative Variance Explained")
15 plt.grid()
16 plt.show()

```

• PCA Loadings Table

This code calculates and displays the loading matrix for all principal components.

```

1 loadings = pca.components_.T * np.sqrt(pca.explained_variance_)
2
3 loadings_df = pd.DataFrame(
4     loadings,
5     index=X_clean.columns,
6     columns=[f"PC{i+1}" for i in range(loadings.shape[1])]
7 )
8
9 print(loadings_df)

```

• Top Contributing Variables

This code identifies the top contributing variables for the first few principal components.

```

1 for i in range(min(6, loadings_df.shape[1])):
2     pc = f"PC{i+1}"
3     print(f"\nTop variables for {pc}:")
4
5     print(loadings_df[pc]
6           .abs()
7           .sort_values(ascending=False)
8           .head(5))

```

- **KMO and Bartlett's Test**

This code checks whether the dataset is suitable for Factor Analysis using the KMO measure and Bartlett's test of sphericity.

```
1 from factor_analyzer.factor_analyzer import calculate_kmo,
   calculate_bartlett_sphericity
2
3 kmo_all, kmo_model = calculate_kmo(X_clean)
4
5 print(f"KMO_Overall_Score: {kmo_model:.3f}")
6
7 chi_square_value, p_value = calculate_bartlett_sphericity(
   X_clean)
8
9 print(f" Bartlett's test p-value: {p_value:.5f}")
```

- **Factor Analysis Model**

This code applies Factor Analysis with 3 factors and varimax rotation.

```
1 from factor_analyzer import FactorAnalyzer
2
3 fa = FactorAnalyzer(n_factors=3, rotation='varimax')
4
5 X_fa = fa.fit_transform(X_scaled)
6
7 print("FA_output_shape:", X_fa.shape)
```

- **Factor Loadings**

This code extracts factor loadings for each variable.

```
1 loadings = pd.DataFrame(
2     fa.loadings_,
3     columns=['F1', 'F2', 'F3'],
4     index=X_clean.columns
5 )
6
7 print(loadings)
```

- **Factor Loadings Heatmap**

This code visualizes factor loadings using a heatmap for better interpretation.

```

1 plt.figure(figsize=(10,6))
2 sns.heatmap(loadings, cmap="coolwarm", center=0)
3 plt.title("Factor_Loadings_Heatmap")
4 plt.show()

```

This code divides the dataset into two variable sets: environmental/demographic features and thermal features.

```

1 set1_cols = [
2     'Gender', 'Age', 'Ethnicity',
3     'T_atm', 'Humidity', 'Distance',
4     'T_offset1'
5 ]
6
7 set2_cols = [
8     'Max1R13_1', 'Max1L13_1',
9     'aveAllR13_1', 'aveAllL13_1',
10    'T_RC1', 'T_RC_Dry1', 'T_RC_Wet1', 'T_RC_Max1',
11    'T_LC1', 'T_LC_Dry1', 'T_LC_Wet1', 'T_LC_Max1',
12    'RCC1', 'LCC1',
13    'canthiMax1', 'canthi4Max1',
14    'T_FHCC1', 'T_FHRC1', 'T_FHLC1', 'T_FHBC1', 'T_FHTC1',
15    'T_FH_Max1', 'T_FHC_Max1',
16    'T_Max1',
17    'T_OR1', 'T_OR_Max1'
18 ]
19
20 X1 = X_clean[set1_cols]
21 X2 = X_clean[set2_cols]

```

• Canonical Correlation Model

This code applies Canonical Correlation Analysis to find relationships between the two variable sets.

```

1 from sklearn.cross_decomposition import CCA
2
3 cca = CCA(n_components=2)
4
5 X1_c, X2_c = cca.fit_transform(X1, X2)

```

- **Canonical Correlations**

This code computes correlation between canonical variates of both datasets.

```
1 canonical_corrs = [  
2     np.corrcoef(X1_c[:, i], X2_c[:, i])[0, 1]  
3     for i in range(2)  
4 ]  
5  
6 print("Canonical_Correlations:")  
7 for i, c in enumerate(canonical_corrs):  
8     print(f"Component_{i+1}:_{c:.4f}")
```

- **Canonical Correlation Bar Plot**

This code visualizes the strength of canonical correlations using a bar chart.

```
1 plt.figure(figsize=(6,4))  
2  
3 plt.bar(['CCA1', 'CCA2'], canonical_corrs)  
4  
5 plt.title("Canonical_Correlations")  
6 plt.xlabel("Components")  
7 plt.ylabel("Correlation")  
8 plt.ylim(0,1)  
9 plt.grid(axis='y', linestyle='--', alpha=0.5)  
10  
11 plt.show()
```

- **CCA Scatter Plot**

This code visualizes the relationship between the first canonical components of both sets.

```
1 plt.figure(figsize=(6,5))  
2  
3 plt.scatter(X1_c[:,0], X2_c[:,0], alpha=0.6)  
4  
5 plt.title("CCA_Scatter_Plot_(Component_1)")  
6 plt.xlabel("Environmental_+_Demographic_Set")  
7 plt.ylabel("Thermal_Feature_Set")  
8  
9 plt.grid(True, linestyle='--', alpha=0.5)
```

```
10 plt.show()
```

- **CCA Correlation Matrix**

This code computes and displays the correlation matrix between canonical components.

```
1 labels = ['X1_CC1', 'X1_CC2', 'X2_CC1', 'X2_CC2']
2
3 cca_corr_df = pd.DataFrame(
4     np.round(np.corrcoef(X1_c.T, X2_c.T), 4),
5     index=labels,
6     columns=labels
7 )
8
9 print(cca_corr_df)
```